

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Debugging

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO2: Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete. (BTL6)

Assessment Tool Weightage: 25%

Question Count: 10

Total Marks: 20

Code of Conduct

I **Yarra Pawan (192424294)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Y.Pawan**

Debugging

Criteria	Problem Identification (4 Marks)	Debugging (4 Marks)	Optimization (4 Marks)	Analysis (4 Marks)	Code Quality (4 Marks)
Marks					

Total Marks Awarded:

Questions:

Question 1: SQLite Database Connection

The following Python program is written to connect to an SQLite database and create a table. However, the program raises an error and does not execute successfully. Analyze the code, identify the mistakes, and rewrite the corrected version.

```
import sqlite3
```

```

conn = sqlite3.connect(student.db)
cursor = conn.cursor
cursor.execute("""
CREATE TABLE Student(
Student_ID INTEGER PRIMARY KEY,
Name TEXT,
Department TEXT
)
""")
conn.commit
conn.close()

```

Question 2: MySQL Record Insertion

The following Python program is intended to connect to a MySQL database and insert a new student record. The program contains multiple errors that prevent successful execution. Debug the code and provide the corrected program.

```

import mysql.connector
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="pawan1611",
    database="college"
)
cursor = conn.cursor()
query = "INSERT INTO Student(Student_ID, Name, Department) VALUES (%s,%s,%s)"
cursor.execute(query, (101, "Alice"))
conn.commit()
conn.close()

```

Question 3: PostgreSQL Data Retrieval

The following Python program is designed to retrieve all records from the Employee table in a PostgreSQL database. The code contains logical and syntax errors. Analyze the program, identify the errors, and write the corrected version.

```

import psycopg2
conn = psycopg2.connect(
    host="localhost",
    database="company",
    user="postgres",
    password="admin"
)
cursor = conn.cursor()
cursor.execute("SELECT * FROM Employee")
rows = cursor.fetchone()

```

```
for row in rows:
    print(row)
cursor.close()
```

Question 4: SQLite Record Update

The following program attempts to update the department of a student in an SQLite database. The update operation fails due to errors in the code. Identify the mistakes and rewrite the corrected program.

```
import sqlite3
conn = sqlite3.connect("college.db")
cursor = conn.cursor()
query = "UPDATE Student SET Department='ECE' WHERE Student_ID=?"
cursor.execute(query)
conn.commit()
conn.close()
```

Question 5: PostgreSQL Record Deletion

The following Python program is intended to delete a customer record from a PostgreSQL database. The code contains errors related to SQL execution and resource management. Debug the program and provide the corrected version.

```
import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="bank",
    user="postgres",
    password="admin"
)

cursor = conn.cursor()

customer_id = 105

cursor.execute("DELETE FROM Customer WHERE Customer_ID=%s")

conn.close()
```

Question 6: SQLite Record Retrieval

The following Python program is intended to retrieve the details of a student whose Student_ID is 105 from an SQLite database. The program produces an error and does not display the required record. Analyze the code, identify the mistakes, and rewrite the corrected version.

```
import sqlite3
```

```

conn = sqlite3.connect("college.db")
cursor = conn.cursor()
student_id = 105
cursor.execute("SELECT * FROM Student WHERE Student_ID=?", student_id)
row = cursor.fetchall()
print(row[0])
conn.close()

```

Question 7: MySQL Table Creation

A student wrote the following Python program to create a table in a MySQL database. The program contains syntax and logical errors. Debug the code and provide the corrected version.

```

import mysql.connector
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="root",
    database="college"
)
cursor = conn.cursor()
cursor.execute("""
CREATE TABLE Student(
Student_ID INT PRIMARY KEY
Name VARCHAR(50),
Department VARCHAR(30)
)
""")
conn.commit()
conn.close()

```

Question 8: PostgreSQL Record Insertion

The following Python program is intended to insert a new employee record into a PostgreSQL database. The program fails during execution due to multiple errors. Analyze the code, identify the mistakes, and rewrite the corrected version.

```

import psycpg2
conn = psycpg2.connect(
    host="localhost",
    database="company",
    user="postgres",
    password="admin"
)

```

```

cursor = conn.cursor()

sql = """
INSERT INTO Employee(Employee_ID, Name, Salary)
VALUES (%s, %s, %s)
"""

cursor.execute(sql, 101, "John", 55000)

conn.commit()

cursor.close()

conn.close()

```

Question 9: MySQL Record Update

The following Python program is intended to update the salary of an employee in a MySQL database. The update operation does not execute successfully. Debug the code and provide the corrected version.

```

import mysql.connector
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="root",
    database="company"
)
cursor = conn.cursor()
salary = 65000
emp_id = 110
query = "UPDATE Employee SET Salary=%s WHERE Employee_ID=%s"
cursor.execute(query, salary, emp_id)
conn.commit()
print("Record Updated")
cursor.close()
conn.close()

```

Question 10: PostgreSQL Record Deletion

The following Python program attempts to delete all customer records belonging to the city Chennai. However, the program contains errors related to parameter passing and transaction handling. Analyze the code, identify the mistakes, and rewrite the corrected version.

```

import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="bank",

```

```
    user="postgres",  
    password="admin"  
)  
  
cursor = conn.cursor()  
  
city = "Chennai"  
  
query = "DELETE FROM Customer WHERE City=%s"  
  
cursor.execute(query)  
  
print("Records Deleted")  
  
cursor.close()  
conn.close()
```

Signature of the Course Faculty

1. SQLite Database Connection

```
import sqlite3
conn = sqlite3.connect("student.db")
cursor = conn.cursor()
cursor.execute("""
CREATE TABLE Student(
    Student_ID INTEGER PRIMARY KEY,
    Name TEXT,
    Department TEXT
)
""")
conn.commit()
conn.close()
```

Errors:

1. "student.db" must be in quotes.
2. conn.cursor → conn.cursor()
3. conn.commit → conn.commit()

2. MySQL Record Insertion

```
import mysql.connector
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="pawan1611",
    database="college"
)
cursor = conn.cursor()
query = "INSERT INTO Student(Student_ID, Name, Department) VALUES (%s, %s, %s)"
cursor.execute(query, (101, "Alice", "CSE"))
conn.commit()
cursor.close()
conn.close()
```

Errors:

1. Three values required but only two were passed.
2. Added Department value "CSE".
3. Closed cursor before closing the connection.

3. PostgreSQL Data Retrieval

```
import psycopg2
conn = psycopg2.connect(
    host="localhost",
```

```

    database="company",
    user="postgres",
    password="pawan1611"
)
cursor = conn.cursor()
cursor.execute("SELECT * FROM Employee")
rows = cursor.fetchall()
for row in rows:
    print(row)
cursor.close()
conn.close()

```

Errors:

1. fetchone() → fetchall()
2. Proper loop indentation.
3. Connection closed.

4. SQLite Record Update

```

import sqlite3
conn = sqlite3.connect("college.db")
cursor = conn.cursor()
query = "UPDATE Student SET Department=? WHERE Student_ID=?"
cursor.execute(query, ("ECE", 101))
conn.commit()
cursor.close()
conn.close()

```

Errors:

1. Used parameterized query.
2. Passed values to execute().
3. Closed cursor.

5. PostgreSQL Record Deletion

```

import psycopg2
conn = psycopg2.connect(
    host="localhost",
    database="bank",
    user="postgres",
    password="pawan1611"
)
cursor = conn.cursor()
customer_id = 105
cursor.execute("DELETE FROM Customer WHERE Customer_ID=%s", (customer_id,))
conn.commit()
cursor.close()

```



```
conn.close()
```

Errors:

1. Passed parameter tuple.
2. Added commit().
3. Closed cursor before closing connection.

6. SQLite Record Retrieval

```
import sqlite3
conn = sqlite3.connect("college.db")
cursor = conn.cursor()
student_id = 105
cursor.execute("SELECT * FROM Student WHERE Student_ID=?", (student_id,))
row = cursor.fetchone()
print(row)
cursor.close()
conn.close()
```

Errors:

1. Parameter must be a tuple (student_id,).
2. Used fetchone() instead of fetchall().
3. Closed cursor.

7. MySQL Table Creation

```
import mysql.connector
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="pawan1611",
    database="college"
)
cursor = conn.cursor()
cursor.execute("""
CREATE TABLE Student(
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Department VARCHAR(30)
)
""")
conn.commit()
cursor.close()
conn.close()
```

Errors:

1. Missing comma after Student_ID INT PRIMARY KEY.

2. Closed cursor.

8. PostgreSQL Record Insertion

```
import psycopg2
conn = psycopg2.connect(
    host="localhost",
    database="company",
    user="postgres",
    password="pawan1611"
)
cursor = conn.cursor()
sql = """INSERT INTO Employee(Employee_ID, Name, Salary)VALUES (%s, %s, %s)"""
cursor.execute(sql, (101, "John", 55000))
conn.commit()
cursor.close()
conn.close()
```

Errors:

1. Parameters must be passed as a tuple.
2. Correct execute() syntax.

9. MySQL Record Update

```
import mysql.connector
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="pawan1611",
    database="company"
)
cursor = conn.cursor()
salary = 65000
emp_id = 110
query = "UPDATE Employee SET Salary=%s WHERE Employee_ID=%s"
cursor.execute(query, (salary, emp_id))
conn.commit()
print("Record Updated")
cursor.close()
conn.close()
```

Errors:

1. Parameters must be passed as a tuple (salary, emp_id).

10. PostgreSQL Record Deletion

```
import psycopg2
conn = psycopg2.connect(
    host="localhost",
    database="bank",
    user="postgres",
    password="pawan1611"
)
cursor = conn.cursor()
city = "Chennai"
query = "DELETE FROM Customer WHERE City=%s"
cursor.execute(query, (city,))
conn.commit()
print("Records Deleted")
cursor.close()
conn.close()
```

Errors:

1. Passed parameter tuple (city,).
2. Added conn.commit().
3. Closed cursor before closing the connection.

<https://colab.research.google.com/drive/1cgrE2wwF-y9tKYxk6Gtx7bh7hW--mtl8?usp=sharing>